SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 6.01

C SAAT Rational Functions, Sequences, Series & Logarithms

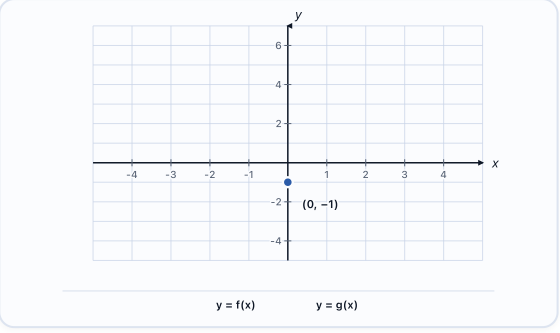
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

SAAT Rational Functions, Sequences, Series & Logarithms Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 16x + 80$ in the domain of SAAT Functions, Sequences & Logarithms. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(10)$.



A 44

B 36

C 20

D 34

2 MCQ

A mathematical model in SAAT Functions, Sequences & Logarithms is defined by the cubic function $y_2(x) = 4x^3 - 8x^2 + 15x - 13$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 123

B 60

C 75

D 35

3 MCQ

In SAAT Functions, Sequences & Logarithms, two displacement vectors are given by $\mathbf{u}_3 = \langle 8, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A 16

B 128

C 256

D 144

4 MCQ

The linear growth profile shown in the diagram for SAAT Functions, Sequences & Logarithms follows an arithmetic progression with first term $a_1 = 9$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 765

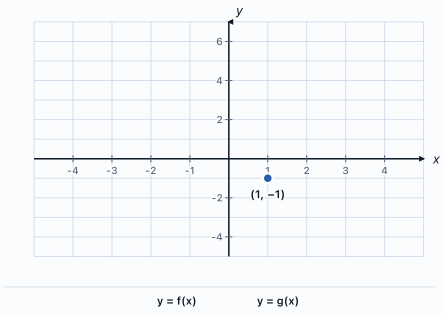
B 93

C 855

D 1395

5 MCQ

In SAAT Functions, Sequences & Logarithms, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 32x)\,\mathrm{d}x$.



A 136

B 112

C 144

D 57

6 MCQ

In SAAT Functions, Sequences & Logarithms, consider the quadratic function $f_6(x) = x^2 - 18x + 169$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(16)$.

A 234

B 137

C 225

D 218

7 MCQ

The graph illustrates the cubic model $y_7(x) = 4x^3 - 9x^2 + 33x - 20$ studied in SAAT Functions, Sequences & Logarithms. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 801

B 624

C 111

D 657

8 MCQ

In SAAT Functions, Sequences & Logarithms, two displacement vectors are given by $\mathbf{u}_8 = \langle 9, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 167

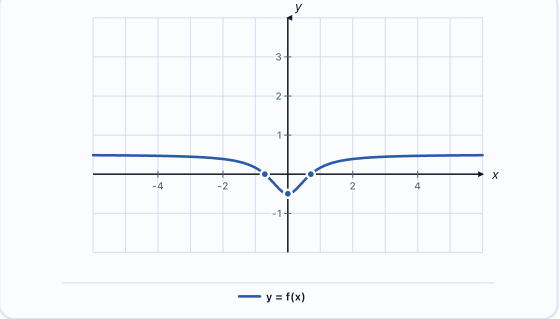
B 37

C 130

D 237

9 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{2x^2-3x+1}{4x^2+2}$.



A $y = 0$

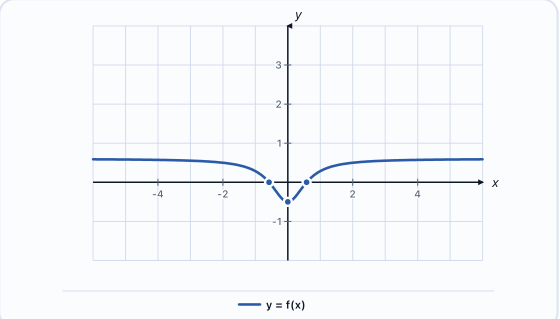
B $y = 2$

C $y = 6$

D $y = \frac{1}{2}$

10 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{3x^3-4x+1}{5x^3+3}$.



A $y = \frac{3}{5}$

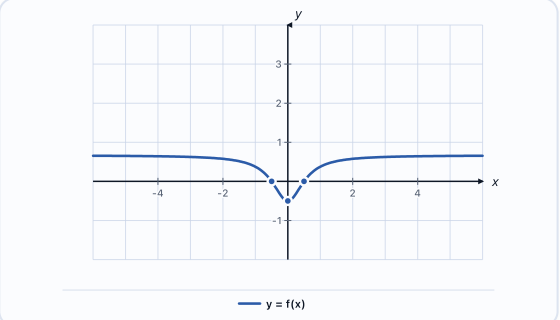
B $y = 0$

C $y = \frac{5}{3}$

D $y = 8$

11 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{4x^4-5x+1}{6x^4+4}$.



A $y = 10$

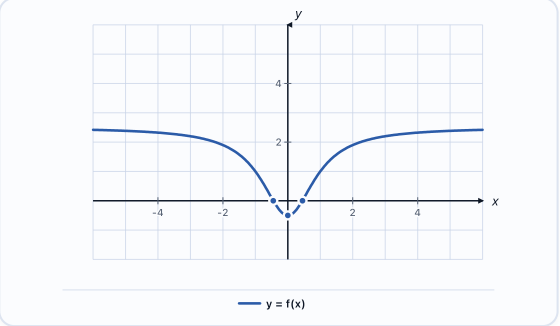
B $y = 0$

C $y = \frac{2}{3}$

D $y = \frac{3}{2}$

12 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{5x^2-6x+1}{2x^2+5}$.



A $y = 0$

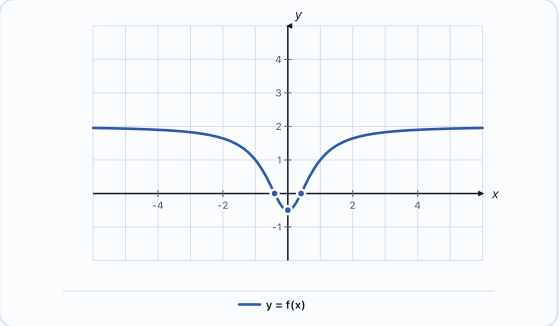
B $y = \frac{5}{2}$

C $y = \frac{2}{5}$

D $y = 7$

13 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{6x^3-7x+1}{3x^3+6}$.



A $y = 0$

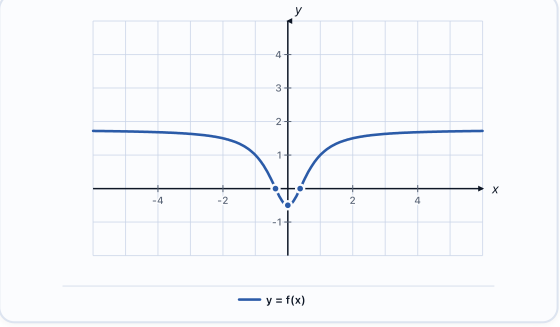
B $y = \frac{1}{2}$

C $y = 2$

D $y = 9$

14 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{7x^4-8x+1}{4x^4+7}$.



A $y = 0$

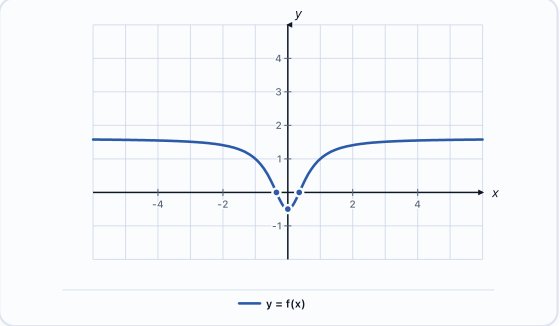
B $y = \frac{7}{4}$

C $y = \frac{4}{7}$

D $y = 11$

15 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{8x^2-9x+1}{5x^2+8}$.



A $y = \frac{8}{5}$

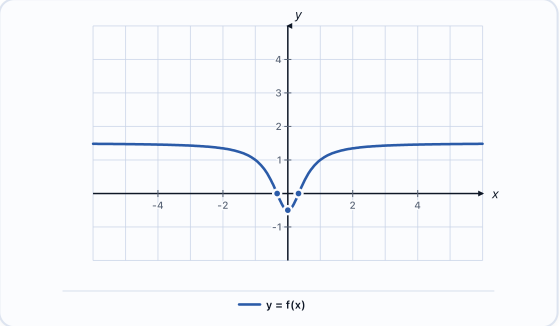
B $y = 13$

C $y = 0$

D $y = \frac{5}{8}$

16 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{9x^3-10x+1}{6x^3+9}$.



A $y = 15$

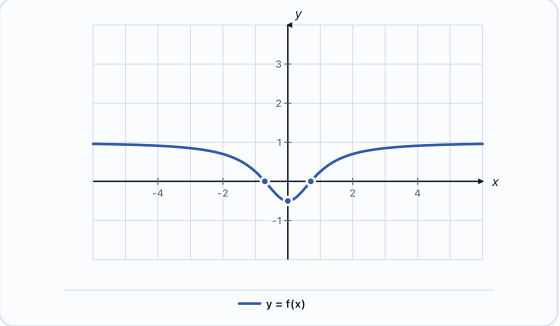
B $y = \frac{3}{2}$

C $y = 0$

D $y = \frac{2}{3}$

17 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{2x^4-11x+1}{2x^4+10}$.



A 14

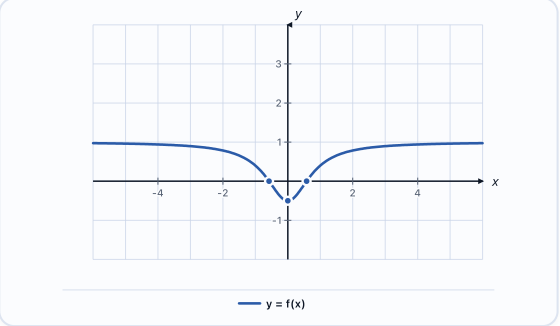
B $y = 0$

C $y = 4$

D $y = 1$

18 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{3x^2-12x+1}{3x^2+11}$.



A 14

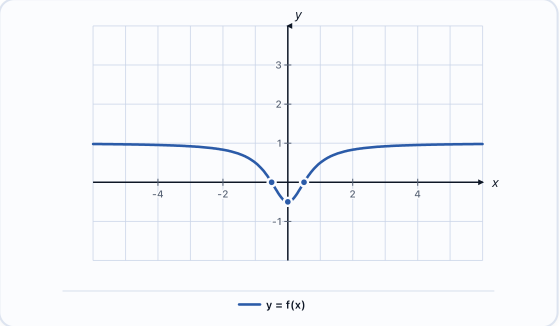
B $y = 1$

C $y = 0$

D $y = 6$

19 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{4x^3-13x+1}{4x^3+12}$.



A $y = 1$

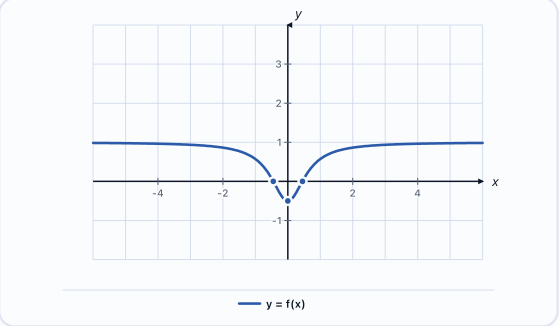
B 14

C $y = 0$

D $y = 8$

20 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{5x^4-14x+1}{5x^4+13}$.



A $y = 10$

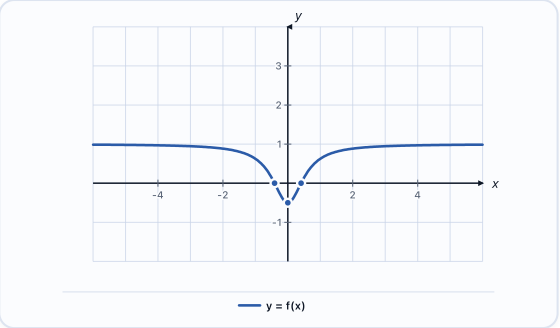
B 14

C $y = 0$

D $y = 1$

21 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{6x^2-15x+1}{6x^2+14}$.



A $y = 0$

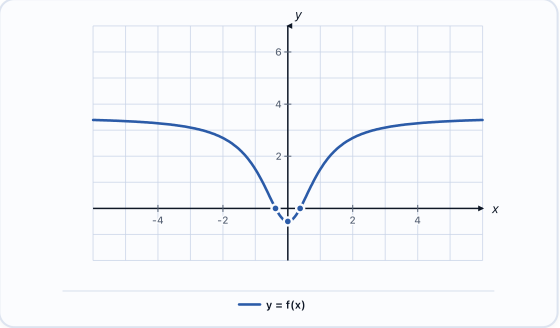
B $y = 1$

C $y = 12$

D 14

22 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{7x^3-16x+1}{2x^3+15}$.



A $y = 0$

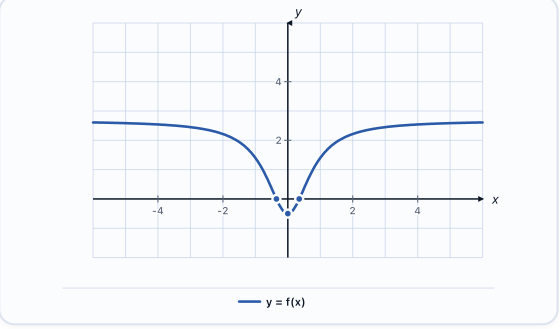
B $y = \frac{2}{7}$

C $y = \frac{7}{2}$

D $y = 9$

23 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{8x^4-17x+1}{3x^4+16}$.



A $y = 11$

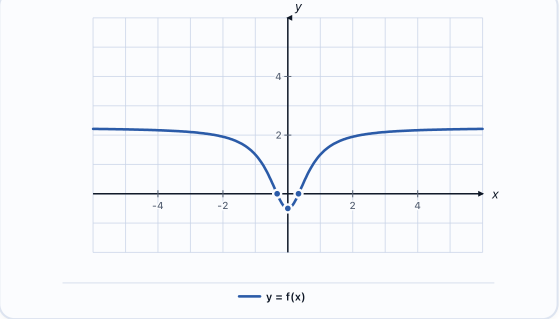
B $y = \frac{3}{8}$

C $y = 0$

D $y = \frac{8}{3}$

24 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{9x^2-18x+1}{4x^2+17}$.



A $y = 13$

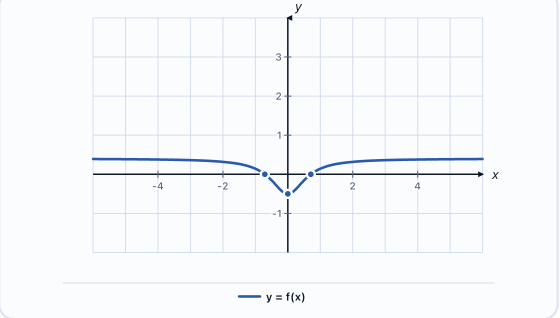
B $y = 0$

C $y = \frac{4}{9}$

D $y = \frac{9}{4}$

25 MCQ

Find the horizontal asymptote of the rational function $f(x) = \frac{2x^3-19x+1}{5x^3+18}$.



A $y = 7$

B $y = \frac{5}{2}$

C $y = 0$

D $y = \frac{2}{5}$